

VetOS Pro

Complete Veterinary Clinic Management System

Project Idea Brief, Feature Plan, UI-First Build Steps, Tech Stack, Database Design, and MVP Scope

Project Positioning: This system is designed for modern veterinary clinics and pet hospitals that need more than a basic POS or appointment register. It manages the full operational journey: appointment request, reception, queue, doctor consultation, medical record, prescription, lab, imaging, pharmacy, inventory, billing handoff, emergency cases, grooming, boarding, staff, reports, and multi-branch control.

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Reference clinic model: PetXpress-style modern veterinary clinic in Lahore. This document uses public PetXpress website information only for general market and feature inspiration.

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Executive Summary

This project is a complete veterinary clinic management system built for modern pet clinics, pet hospitals, and multi-branch veterinary businesses. The system is not only an appointment or billing tool. It is a complete operational platform that connects reception, doctors, lab, imaging, pharmacy, inventory, billing handoff, emergency workflow, grooming, boarding, staff, and reports in one place. The customer side will stay simple in the first version: a website with a form, call button, WhatsApp button, branch information, service pages, and emergency contact. The internal clinic side will be powerful: reception can receive appointment requests, confirm them manually, add pets to the queue, and track their full journey inside the clinic. Doctors can open a pet's complete medical record, write SOAP notes, create prescriptions, order lab tests, request imaging, record vaccination and deworming, and schedule follow-ups. Lab and imaging staff can process orders and upload reports. Pharmacy and inventory staff can dispense medicines and track stock, batch numbers, expiry dates, and suppliers. Admin can view revenue, appointments, staff performance, branch comparison, low stock, expiring medicines, pending lab reports, and overall clinic performance.

The best initial strategy is to build the system UI first. The first demo should show a complete realistic clinic workflow: a customer submits an appointment form, reception confirms it, the pet is checked in, the queue is updated, the doctor performs consultation, lab tests are ordered, pharmacy dispenses medicines, billing summary is generated for POS handoff, and the visit is saved in the pet's medical timeline. This makes the product feel complete and sellable even before advanced features such as customer portal, mobile app, AI summaries, and direct POS API integration are added.

Verified Market Reference: PetXpress-Style Clinic

The system is inspired by the workflow needs of a modern Lahore veterinary clinic. Public pages on the PetXpress website list services such as diagnostics, vaccination, deworming, surgeries, dentistry, endoscopy, clinical examination, emergency care, doctors, and multiple branches such as DHA, Lake City, UVAS, and GT Road. Their branch pages also show appointment forms and call or WhatsApp calls to action. This supports the decision to keep the MVP customer-side appointment process simple: form, call, and WhatsApp instead of full self-booking in the first version.

Important Product Decision: The MVP will not include loyalty or membership plans. It will also not include full online self-booking, mobile app, customer login, or online payment in the first version. The first version will use a simple appointment request form plus call and WhatsApp buttons, while the full appointment workflow will be handled internally by reception.

Reference Sources

- PetXpress official website: <https://petxpress.pk/>
- PetXpress emergency page: <https://petxpress.pk/emergency>
- PetXpress DHA branch page: <https://petxpress.pk/pet-clinic-in-dha>
- PetXpress Lake City branch page: <https://petxpress.pk/pet-clinic-lake-city>
- PetXpress diagnostics page: <https://petxpress.pk/diagnostics>

Product Vision

Product Name

Recommended name:

VetOS Pro

Product Tagline

A complete operating system for modern veterinary clinics, pet hospitals, grooming centers, boarding facilities, and multi-branch pet-care businesses.

One-Line Pitch

A complete veterinary clinic management system that covers appointment requests, reception, queue, medical records, prescriptions, lab, imaging, pharmacy, inventory, billing handoff, emergency cases, grooming, boarding, staff, reports, and multi-branch operations.

Main Selling Angle

Do not pitch this as simple appointment software. Pitch it as a workflow and operations platform:

Pitch: Your website form, reception, doctor consultation, lab, pharmacy, inventory, and billing workflow will be connected in one system, so no patient, medicine, report, or follow-up gets lost.

Target Users and Roles

Role	Main Responsibilities
Super Admin	Full system control, branches, users, pricing, reports, settings, audit logs.
Branch Admin	Manage one branch, staff, doctors, revenue, queue, branch reports.
Receptionist	Appointment requests, client registration, pet registration, check-in, queue, follow-up calls, billing handoff.
Veterinarian	Consultation, EMR, SOAP notes, diagnosis, prescription, lab orders, imaging requests, vaccination, deworming, follow-ups.
Lab Staff	Receive lab orders, collect samples, enter results, upload reports, mark critical results.
Imaging Staff	Process X-ray, ultrasound, endoscopy, and imaging requests; upload reports and images.
Pharmacy Staff	Receive prescriptions, dispense medicines, update stock, handle substitutes and partial dispensing.

Inventory Manager	Manage stock, batches, expiry dates, suppliers, purchase orders, transfers, adjustments.
Grooming Staff	Grooming appointments, package details, before/after photos, behavior notes.
Boarding Staff	Boarding check-ins, kennel assignment, feeding, medication, daily pet updates.
Accountant/Billing Staff	Review bill summaries, manage payment status, export billing data, POS handoff.

System Architecture Overview

The product has three major parts:

1. Internal Clinic Management System

This is the main dashboard application used by staff. It contains the admin dashboard, reception workflow, doctor workflow, lab, imaging, pharmacy, inventory, billing handoff, reports, grooming, boarding, and emergency handling.

2. Public Website

This is the customer-facing website. It has clinic information, services, doctors, branches, emergency contact, appointment form, call button, and WhatsApp button. It does not need customer login in the MVP.

3. Future Customer Portal or Mobile App

This is not part of the first MVP. Later, pet owners can log in to view prescriptions, lab reports, vaccination cards, invoices, pet profiles, boarding updates, and appointment history.

MVP Appointment Strategy

Why Simple Form, Call, and WhatsApp First

For the first version, full online slot booking is unnecessary and can make the build too complex. Veterinary clinics often prefer manual confirmation because emergencies, doctor availability, service duration, surgery timing, and pet condition can change the schedule. A form plus call and WhatsApp is realistic, easy to build, and close to what many clinics already use.

Customer-Side Flow

1. Customer opens the website.
2. Customer clicks **Book Appointment**, **Call Now**, or **WhatsApp Now**.
3. If the form is used, customer enters owner details, pet details, branch, service, preferred date/time, and complaint.
4. System shows: *Your appointment request has been received. Our team will contact you shortly.*

Clinic-Side Flow

1. Reception receives the appointment request in the dashboard.
2. Reception calls or WhatsApps the owner.
3. Reception confirms, reschedules, cancels, or rejects the request.
4. Confirmed request becomes an official appointment.
5. On visit day, pet is checked in and added to queue.
6. Doctor starts consultation.

Detailed Feature Modules

Module 1: Admin Dashboard

The admin dashboard gives the clinic owner or manager a full overview of operations. It should immediately show today's revenue, appointments, emergency cases, active queue, pending lab reports, pending pharmacy requests, low-stock medicines, expiring medicines, new clients, repeat clients, and branch performance.

Functions

- View today's revenue and monthly revenue.
- View confirmed, completed, cancelled, and no-show appointments.
- View emergency cases and their status.
- View active queue and waiting time.
- View pending lab, imaging, pharmacy, and billing tasks.
- View low-stock and near-expiry medicines.
- View branch-wise and doctor-wise performance.
- Export reports as PDF or Excel.
- Filter reports by branch, doctor, service, and date range.

Module 2: Branch Management

This module supports clinics with multiple branches. Each branch can have its own doctors, staff, services, appointments, inventory, billing, and reports, while the central admin can monitor all branches.

Functions

- Add, edit, and disable branches.
- Store branch name, address, phone, WhatsApp, timings, and emergency availability.
- Assign doctors and staff to branches.
- Enable or disable services per branch.
- View branch-specific appointment calendar.

- Track branch inventory and medicine stock.
- Compare revenue and performance across branches.
- Transfer pet record between branches.

Module 3: User Roles and Permissions

The system must have strict role-based access. Reception should not have the same controls as doctors, and lab staff should not edit billing settings.

Functions

- Create roles such as admin, receptionist, doctor, lab staff, pharmacy staff, inventory manager, groomer, boarding staff, and accountant.
- Assign permissions for viewing, creating, editing, deleting, exporting, and approving data.
- Restrict users by branch.
- Disable inactive users.
- Track user activity through audit logs.

Module 4: Client Management

This module stores pet owner information. The most important search field should be phone or WhatsApp number because reception commonly identifies customers by phone.

Client Fields

- Name, phone, WhatsApp, email, address, city.
- Emergency contact.
- Preferred branch and preferred doctor.
- Client tags: VIP, breeder, rescue case, regular client, difficult client, credit allowed.
- Total pets, total visits, total spending, outstanding balance.
- Internal notes and communication history.

Functions

- Add new client.
- Search client by phone number.
- Edit client details.
- Add pet under client.
- View appointment, invoice, and visit history.
- Detect duplicate clients by phone number.
- Call or WhatsApp client directly from profile.

Module 5: Pet Management

This is one of the most important modules. Every pet should have a complete medical and service profile.

Pet Fields

- Pet name, species, breed, gender, age/date of birth, color, photo.
- Owner information.
- Weight history.
- Microchip number if available.
- Neutered/spayed status.
- Allergies, chronic diseases, current medicines.
- Temperament, aggression warning, bite history, special handling notes.
- Food preference.
- Vaccination and deworming status.
- Last visit and next follow-up.

Functions

- Add and edit pet profile.
- Upload pet photo.
- Add allergy and medical warnings.
- Add weight record and view weight graph later.
- View medical history, prescriptions, lab reports, imaging reports, vaccinations, deworming, grooming, boarding, and invoices.
- Show pet medical timeline.

Advanced Feature: Pet Medical Timeline

The pet profile should include a timeline showing every important event: appointment, consultation, vaccination, deworming, lab test, imaging, prescription, surgery, grooming, boarding, invoice, and follow-up. This gives doctors a quick view of the pet's full history.

Module 6: Appointment Request Inbox

This module receives appointment requests from the public website form. It is not a full self-booking system; reception confirms manually.

Request Fields

- Owner name, phone, WhatsApp, email.
- Pet name, species, breed optional.
- Service required.
- Preferred branch.

- Preferred date and time.
- Complaint or additional details.
- Emergency checkbox.

Functions

- View new requests.
- Mark as contacted.
- Confirm, reschedule, cancel, or reject request.
- Convert request into official appointment.
- Create client and pet profile from request.
- Detect duplicate requests.

Module 7: Appointment Calendar

Confirmed appointments are managed in the appointment calendar. Reception and doctors should be able to view appointments by date, doctor, branch, and service type.

Appointment Types

- Consultation, vaccination, deworming, follow-up.
- Diagnostics, X-ray, ultrasound, endoscopy.
- Surgery, dentistry, emergency.
- Grooming, boarding, home visit, online consultation optional.

Statuses

- Confirmed, checked in, waiting, in consultation.
- Lab pending, imaging pending, pharmacy pending, billing pending.
- Completed, cancelled, no-show.

Functions

- Create appointment manually.
- Edit appointment.
- Assign doctor, branch, room, and service.
- Mark checked in and add to queue.
- Mark no-show or cancelled.
- Print appointment slip if needed.

Module 8: Reception Queue System

The queue system controls what happens after the pet arrives. It prevents confusion between reception, doctor rooms, lab, pharmacy, and billing.

Queue Fields

- Token number, pet name, owner name.
- Doctor, service, branch, arrival time.
- Priority level and waiting time.
- Current status and internal notes.

Queue Statuses

- Waiting at reception.
- Waiting for doctor.
- In consultation.
- Lab pending.
- Imaging pending.
- Pharmacy pending.
- Billing pending.
- Completed.

Module 9: Emergency Case Management

Emergency cases should bypass normal queue when needed. The system should support fast entry and triage.

Functions

- Emergency fast check-in with minimum details.
- Assign triage level: critical, urgent, stable, observation, discharged.
- Record arrival time, complaint, vitals, doctor, immediate treatment, consent, estimated cost, and outcome.
- Show active emergency dashboard.
- Mark emergency cases with priority badges in queue.

Module 10: Doctor Dashboard

The doctor dashboard should make consultation fast and organized.

Functions

- View today's appointments and waiting patients.
- View emergency cases assigned to the doctor.
- View follow-ups due.
- View lab reports pending review.
- Start consultation from queue.
- Open pet profile and medical history.

Module 11: Electronic Medical Record

The EMR stores the complete medical visit record. It is the strongest professional feature of the system.

Consultation Sections

- Pet summary and owner details.
- Presenting complaint.
- Vitals and physical examination.
- Diagnosis and differential diagnosis.
- Treatment plan.
- Prescription.
- Lab and imaging orders.
- Follow-up and owner instructions.

SOAP Format

- **Subjective:** Owner complaint, symptoms, duration, appetite, vomiting, diarrhea, coughing, lethargy, injury history, behavior change.
- **Objective:** Temperature, pulse, respiration, weight, hydration, gum color, pain score, body condition score, physical exam findings.
- **Assessment:** Primary diagnosis, differential diagnosis, severity, risk level.
- **Plan:** Medicines, lab tests, imaging, surgery, diet advice, follow-up date, home care instructions.

Module 12: Clinical Templates

Templates reduce repetitive typing for doctors.

Templates to Include

- Vomiting, diarrhea, fever, skin allergy, ear infection.
- Parvo suspected, tick fever, cat flu, kennel cough.
- Wound case, dental case, vaccination visit, deworming visit.
- Surgery follow-up and post-operative checkup.

Functions

- Use default template.
- Edit template during consultation.
- Save custom doctor template.
- Create branch-level templates.

Module 13: Prescription Management

Doctor creates digital prescriptions and pharmacy dispenses from them.

Prescription Fields

- Medicine name, strength, dose, frequency, duration, route.
- Quantity, refill allowed, instructions, doctor notes.
- Doctor signature and branch details.

Functions

- Create prescription.
- Use prescription templates.
- Send to pharmacy queue.
- Print or export prescription PDF.
- Add home care and diet advice.
- Track dispensed, partially dispensed, or unavailable medicines.

Module 14: Vaccination Management

Vaccination records should be connected to pet profile, medicine inventory, and reminders.

Fields

- Vaccine name, brand, batch number, expiry date.
- Dose, route, date given, doctor, next due date.

Functions

- Add vaccination record.
- Generate vaccination certificate or card.
- Set next due date.
- Track vaccine batch and expiry.
- Send reminder later through WhatsApp/SMS module.

Module 15: Deworming Management

Deworming should be handled like a recurring medical record.

Functions

- Add deworming record.
- Store medicine, dose, weight, date, and next due date.
- Add reminder.
- Show deworming history in pet profile.

Module 16: Lab and Diagnostics

This module manages lab tests from doctor order to final report.

Lab Test Types

- CBC, LFT, RFT, blood glucose.
- Urine test, stool test, skin scraping.
- Parvo, distemper, tick fever, culture test.
- Custom tests.

Workflow

1. Doctor orders test.
2. Lab receives order.
3. Sample is collected.
4. Test is marked in progress.
5. Result is entered.
6. Report is uploaded or generated.
7. Doctor reviews result.
8. Report is attached to pet record.

Advanced Functions

- Abnormal result highlighting.
- Reference range fields.
- Previous result comparison.
- Critical result alert.
- PDF lab report generation.

Module 17: Imaging Management

This module handles X-ray, ultrasound, endoscopy, dental imaging, and other reports.

Functions

- Doctor requests imaging.
- Imaging staff receives request.
- Upload image/report.
- Add findings and impression.
- Doctor reviews report.
- Attach report to pet medical timeline.

Module 18: Surgery and Procedure Management

This module handles planned and emergency procedures.

Functions

- Book surgery or procedure.
- Assign surgeon and assistant.
- Add pre-op checklist.
- Add anesthesia plan.
- Generate consent form.
- Add operation notes.
- Add medicines used.
- Add post-op monitoring.
- Generate discharge instructions.
- Schedule follow-up.

Module 19: Dentistry Management

Dentistry should have a separate workflow for dental exams, scaling, extractions, and follow-ups.

Functions

- Dental examination record.
- Gum and teeth condition notes.
- Dental images.
- Scaling/extraction procedure notes.
- Dental invoice item.
- Dental follow-up reminder.
- Home care instructions.

Module 20: Pharmacy Management

Pharmacy receives prescriptions, dispenses medicines, and updates stock.

Workflow

1. Doctor creates prescription.
2. Pharmacy receives request.
3. Pharmacy checks availability.
4. Pharmacy dispenses medicine.
5. Stock decreases.

6. Billing item is created.
7. Prescription status becomes dispensed or partially dispensed.

Functions

- Pharmacy queue.
- Medicine availability check.
- Substitute medicine request.
- Partial dispensing.
- Out-of-stock marking.
- Medicine label printing later.

Module 21: Inventory Management

Inventory controls medicines, vaccines, lab supplies, surgery supplies, grooming products, food, and accessories.

Fields

- Item name, category, brand, generic name.
- Unit, purchase price, sale price.
- Supplier, branch, storage condition.
- Batch number, expiry date, quantity, minimum stock level.

Functions

- Add, edit, and disable inventory item.
- Stock in, stock out, stock adjustment.
- Transfer stock between branches.
- Batch tracking.
- Expiry tracking.
- Low-stock alert.
- Supplier purchase history.
- Purchase order generation.
- Remove damaged or expired stock.

Module 22: Billing Summary and POS Handoff

Because the clinic may already have POS, the MVP should not replace it. It should create a medical bill summary and pass it to billing staff or POS operator.

Billable Items

- Consultation, emergency fee, vaccination, deworming.
- Lab tests, imaging, surgery, dentistry.
- Grooming, boarding, medicines, supplies, miscellaneous charges.

MVP Billing Flow

1. Doctor/lab/pharmacy adds billable items.
2. System creates visit bill summary.
3. Reception or accountant reviews summary.
4. POS operator enters amount in existing POS.
5. Payment status is manually marked paid, pending, or partial.

Future POS Integration Levels

- Level 1: Manual billing handoff.
- Level 2: CSV/Excel export.
- Level 3: API sync if existing POS supports it.

Module 23: Grooming Management

This optional module manages grooming appointments and records.

Functions

- Grooming appointment.
- Groomer assignment.
- Service package selection.
- Bath, haircut, nail trimming, ear cleaning, tick/flea bath, medicated bath.
- Before/after photos.
- Pet behavior notes.
- Owner grooming preferences.
- Skin sensitivity and aggression notes.

Module 24: Boarding and Daycare Management

This optional module manages pets staying at the clinic.

Functions

- Boarding booking.
- Check-in and check-out.
- Kennel or room assignment.

- Feeding schedule.
- Medication schedule.
- Walk schedule.
- Cleaning status.
- Daily health notes.
- Owner update message.
- Boarding invoice item.

Module 25: Consent Forms

Consent forms are required for surgeries, anesthesia, emergency treatment, grooming, boarding, and high-risk procedures.

Functions

- Create consent form from template.
- Fill owner, pet, procedure, risk, and cost details.
- Capture digital signature later.
- Generate PDF.
- Attach signed form to pet record.

Module 26: Communication System

For MVP, communication can remain semi-manual using WhatsApp and call buttons. Automation can come later.

MVP Functions

- Call button from client profile.
- WhatsApp button from client profile.
- Copy message templates.
- Log communication manually.

Future Automation

- Appointment confirmation.
- Appointment reminder.
- Vaccination reminder.
- Deworming reminder.
- Follow-up reminder.
- Lab report ready message.
- Prescription ready message.

- Grooming completed message.
- Boarding update message.

Module 27: Reports and Analytics

Reports help clinic owners make better decisions.

Business Reports

- Daily, weekly, monthly revenue.
- Branch revenue.
- Doctor revenue.
- Service revenue.
- Pending payments.
- Appointment no-shows.
- Client growth.
- Repeat clients.

Medical Reports

- Most common diseases.
- Vaccination count.
- Deworming count.
- Lab test count.
- Surgery count.
- Emergency case count.
- Follow-up compliance.

Inventory Reports

- Low-stock items.
- Expired items.
- Near-expiry items.
- Fast-moving medicines.
- Slow-moving medicines.
- Supplier purchase report.
- Stock transfer report.

Module 28: Staff Management

Admin manages employees, roles, branches, and schedules.

Functions

- Add staff profile.
- Assign role and branch.
- Set working hours.
- Set doctor availability.
- Track active/inactive status.
- View staff activity.

Module 29: Doctor Schedule Management

Doctors may work in different branches and timings. Reception needs this information to assign appointments correctly.

Functions

- Create doctor schedule.
- Assign branch, days, timings, service types.
- Mark leave or unavailable slots.
- Assign emergency duty.
- Assign surgery slots.

Module 30: Public Website

The website is the public face of the clinic. It should look modern and trustworthy.

Pages

- Home page.
- Services page.
- Branches page.
- Doctors page.
- Appointment form page.
- Emergency page.
- Contact page.

Home Page Sections

- Hero section with book appointment, call, WhatsApp, emergency buttons.
- Services overview.
- Branch cards.
- Doctor highlights.
- Why choose us.

- Reviews section.
- Contact information.

Module 31: Future Customer Portal

This is not part of MVP but should be planned in architecture.

Future Functions

- Phone OTP login.
- View pets.
- View prescriptions.
- View lab reports.
- View vaccination card.
- View invoices.
- View appointment history.
- View boarding updates.

Module 32: AI Features for Later

AI features should assist documentation, not diagnose pets independently.

Safe AI Features

- AI SOAP note drafting.
- AI pet history summary.
- AI discharge summary draft.
- AI follow-up message draft.
- AI owner education text.
- AI lab report summarization for doctor review.

Safety Rule: AI should support documentation only. Final diagnosis, treatment, prescription, and medical decision must remain with the veterinarian.

Module 33: Audit Logs and Security

Medical and billing software needs accountability.

Audit Logs Track

- Who created, edited, deleted, or exported records.
- Who changed appointment, invoice, payment, prescription, or stock status.
- Who viewed sensitive pet/client records.
- Date and time of each action.

Security Features

- Role-based permissions.
- Branch-based access.
- Strong passwords.
- Session timeout.
- Data backup.
- Admin-only settings.

Module 34: Settings Module

The settings module lets admin customize the system.

Settings

- Clinic name and logo.
- Branches.
- Services and pricing.
- Doctors and staff.
- Medicine categories.
- Lab test categories.
- Appointment statuses.
- Prescription templates.
- Invoice template.
- Report template.
- Message templates.
- Currency and tax settings optional.

MVP Scope

MVP Must Include

1. Login and role system.
2. Admin dashboard.
3. Basic branch management.
4. Reception dashboard.
5. Client management.
6. Pet management.
7. Appointment request inbox.

8. Appointment calendar.
9. Queue system.
10. Emergency fast check-in.
11. Doctor dashboard.
12. EMR consultation screen.
13. SOAP notes.
14. Prescription management.
15. Vaccination records.
16. Deworming records.
17. Lab order and report upload.
18. Imaging request and report upload.
19. Basic inventory.
20. Pharmacy queue.
21. Billing summary/POS handoff.
22. Basic reports.
23. Public website.
24. Appointment form.
25. Call and WhatsApp CTA buttons.

Not Included in MVP

- Loyalty and membership plans.
- Full mobile app.
- Customer login portal.
- Real-time online self-booking.
- Online payment.
- Full POS replacement.
- Complex accounting.
- AI diagnosis.
- Insurance module.
- Too much automation before core workflow is stable.

UI-First Build Plan

Step 1: Design System

Build reusable components first:

- Buttons, inputs, dropdowns, date picker.
- Cards, tables, badges, tabs, modals.
- Sidebar, topbar, breadcrumbs.
- Empty states, loading states, error states.
- PDF preview card, file upload card, pet avatar card.

Step 2: Main Layouts

Create role-based layouts:

- Admin layout.
- Reception layout.
- Doctor layout.
- Lab layout.
- Pharmacy layout.
- Inventory layout.
- Public website layout.

Step 3: Core Screens

Build these screens with dummy data:

- Login page.
- Admin dashboard.
- Reception dashboard.
- Appointment request inbox.
- Appointment calendar.
- Queue board.
- Client list.
- Client profile.
- Pet profile.
- Doctor dashboard.
- Consultation screen.
- Prescription screen.

- Lab dashboard.
- Imaging report screen.
- Pharmacy queue.
- Inventory list.
- Billing summary screen.
- Public website home page.
- Public appointment form.

Step 4: Clickable Demo Flow

The first demo should show a complete visit:

1. Website appointment form submitted.
2. Reception receives appointment request.
3. Reception confirms appointment.
4. Pet checked in.
5. Pet added to queue.
6. Doctor opens pet profile.
7. Doctor creates SOAP note.
8. Doctor creates prescription.
9. Doctor orders lab test.
10. Lab uploads report.
11. Pharmacy dispenses medicine.
12. Billing summary generated.
13. Visit completed and saved in pet timeline.

Recommended Tech Stack

Layer	Recommended Technology	Tech-	Reason
Frontend	React + Vite		Fast, modern, suitable for dashboards and SaaS UI.
Styling/UI	Tailwind CSS + shadcn/ui		Clean dashboard components, reusable UI, professional look.
Backend	Django + Django REST Framework		Strong Python backend, secure, good for business systems and admin panel.

Database	PostgreSQL	Reliable relational database for medical, billing, inventory, and branch data.
Authentication	JWT or session auth with role permissions	Required for secure staff login and role-based access.
File Storage MVP	Local file storage	Easy for first version.
File Storage Production	AWS S3 or Cloudflare R2	Better for lab reports, imaging, pet photos, consent forms, prescriptions.
PDF Generation	WeasyPrint / ReportLab / wkhtmltopdf	For prescriptions, invoices, lab reports, vaccination cards, consent forms.
Notifications MVP	Manual WhatsApp/call buttons	Simple and realistic for first build.
Notifications Later	WhatsApp Business API/SMS gateway	Automated reminders and reports later.
Deployment MVP	Local server or VPS	Easier demo and clinic deployment.
Deployment Production	VPS/cloud with backups	More reliable for multi-branch clinic operations.

Database Design Overview

Core Tables

- users, roles, permissions, branches, staff, doctor_schedules.
- clients, pets, pet_weight_history, pet_allergies, pet_medical_warnings.
- appointment_requests, appointments, queue_entries, emergency_cases.

Medical Tables

- visits, consultation_notes, diagnoses, prescriptions, prescription_items.
- vaccines, vaccination_records, deworming_records, followups.
- lab_tests, lab_orders, lab_results.
- imaging_orders, imaging_reports.
- surgeries, surgery_notes, consent_forms.

Inventory and Billing Tables

- inventory_items, medicine_batches, stock_movements.
- suppliers, purchase_orders, pharmacy_dispenses.
- bill_summaries, bill_items, payments, pos_sync_logs.

Extra Services and Logs

- grooming_appointments, boarding_bookings, kennel_rooms, boarding_daily_updates.

- message_templates, communication_logs.
- audit_logs, staff_activity_logs.

Suggested Folder Structure

Frontend Folder Structure

```
frontend/  
  src/  
    app/  
    components/  
      ui/  
      layout/  
      cards/  
      tables/  
      forms/  
    pages/  
      auth/  
      admin/  
      reception/  
      doctor/  
      lab/  
      pharmacy/  
      inventory/  
      public-site/  
    features/  
      clients/  
      pets/  
      appointments/  
      queue/  
      consultations/  
      prescriptions/  
      lab/  
      imaging/  
      pharmacy/  
      inventory/  
      billing/  
      reports/  
    services/  
    hooks/  
    utils/  
    data/
```

Backend Folder Structure

```
backend/  
  config/  
  apps/  
    accounts/  
    branches/  
    clients/  
    pets/
```

```
appointments/  
queue/  
emergency/  
consultations/  
prescriptions/  
vaccination/  
lab/  
imaging/  
pharmacy/  
inventory/  
billing/  
grooming/  
boarding/  
reports/  
communications/  
audit/  
media/  
static/  
requirements.txt
```

API Design Examples

Endpoint	Method	Purpose
/api/auth/login	POST	Staff login.
/api/clients	GET/POST	List or create clients.
/api/clients/id	GET/PATCH	View or update client profile.
/api/pets	GET/POST	List or create pets.
/api/pets/id/timeline	GET	View pet medical timeline.
/api/appointment-requests	GET/POST	Receive website appointment requests.
/api/appointment-requests/id/confirm	POST	Convert request to confirmed appointment.
/api/appointments	GET/POST	Manage appointments.
/api/queue	GET/POST	Manage live queue.
/api/visits	GET/POST	Create consultation visit.
/api/prescriptions	GET/POST	Create prescriptions.
/api/lab-orders	GET/POST	Create and view lab orders.
/api/imaging-orders	GET/POST	Create and view imaging requests.
/api/pharmacy/queue	GET	View pharmacy queue.
/api/inventory/items	GET/POST	Manage inventory items.
/api/billing/summaries	GET/POST	Generate bill summaries.
/api/reports/dashboard	GET	Admin dashboard metrics.

Development Roadmap

Phase 1: Planning and UI Design

- Finalize modules and MVP scope.
- Create sitemap and user flows.
- Create design system.
- Build UI prototype with dummy data.
- Prepare demo story.

Phase 2: Backend Foundation

- Setup Django project.
- Setup PostgreSQL.
- Add authentication and roles.
- Add branches, users, staff.
- Create core database models.

Phase 3: Reception and Appointment Workflow

- Public appointment form.
- Appointment request inbox.
- Client and pet creation.
- Appointment calendar.
- Queue system.
- Emergency fast check-in.

Phase 4: Doctor and EMR Workflow

- Doctor dashboard.
- Pet medical timeline.
- Consultation screen.
- SOAP notes.
- Prescription.
- Vaccination and deworming.
- Follow-up scheduling.

Phase 5: Lab, Imaging, Pharmacy, Inventory

- Lab order workflow.
- Lab result upload.
- Imaging request workflow.
- Pharmacy queue.

- Inventory item, batch, expiry, and stock management.

Phase 6: Billing, Reports, and Demo Polish

- Billing summary.
- POS handoff status.
- Basic reports.
- Dashboard polish.
- PDF generation.
- Demo data and presentation script.

Demo Script for Selling

A pet owner submits a website appointment form because their dog is vomiting. The request appears instantly in the reception dashboard. Reception calls the owner, confirms the appointment, assigns a doctor, and adds the appointment to the calendar. When the owner arrives, reception checks in the pet and adds it to the live queue. The doctor opens the pet profile, sees previous vaccination history and allergy warnings, writes a SOAP consultation note, orders a CBC and parvo test, and creates a prescription. The lab receives the order, uploads results, and marks abnormal values. The doctor reviews the report and schedules a follow-up. Pharmacy dispenses medicines and stock is reduced. Finally, a billing summary is generated for POS handoff. The whole visit is saved in the pet's medical timeline.

Final Build Recommendation

Build the first version as a professional internal clinic management system with a simple customer-facing website form. Do not build the full app, customer portal, loyalty system, online payment, or direct POS replacement in the MVP. Focus on the workflow that will convince a clinic owner: reception, queue, doctor consultation, pet medical history, prescriptions, lab, imaging, pharmacy, inventory, billing handoff, emergency handling, and reports. After the clinic accepts the system, upgrade it with customer portal, WhatsApp automation, AI documentation assistant, POS API integration, and mobile app.